XO-1b

R-1171

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_XO-1_210512` · created 2026-07-31T03:21:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459342.7917 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.033
Transit depth (Rp/Rs)^2	1.28 +/- 0.74 [%]
Orbital Inclination (inc)	89.71 +/- 1.56
Airmass coefficient 1 (a1)	1.3717 +/- 0.0096
Airmass coefficient 2 (a2)	-0.173 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	None
Optimal Aperture	4.27
Optimal Annulus	5.49
Transit Duration (day)	0.0527 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.113 ± 0.033 vs 0.1326 (deviation 0.58σ)
Duration fit vs published	0.0527 d vs 0.1238 d (deviation 11.79σ)
Mid-time O–C	-5674.2 min
Depth SNR	3.4
Residual scatter	0.7 %

Light curve

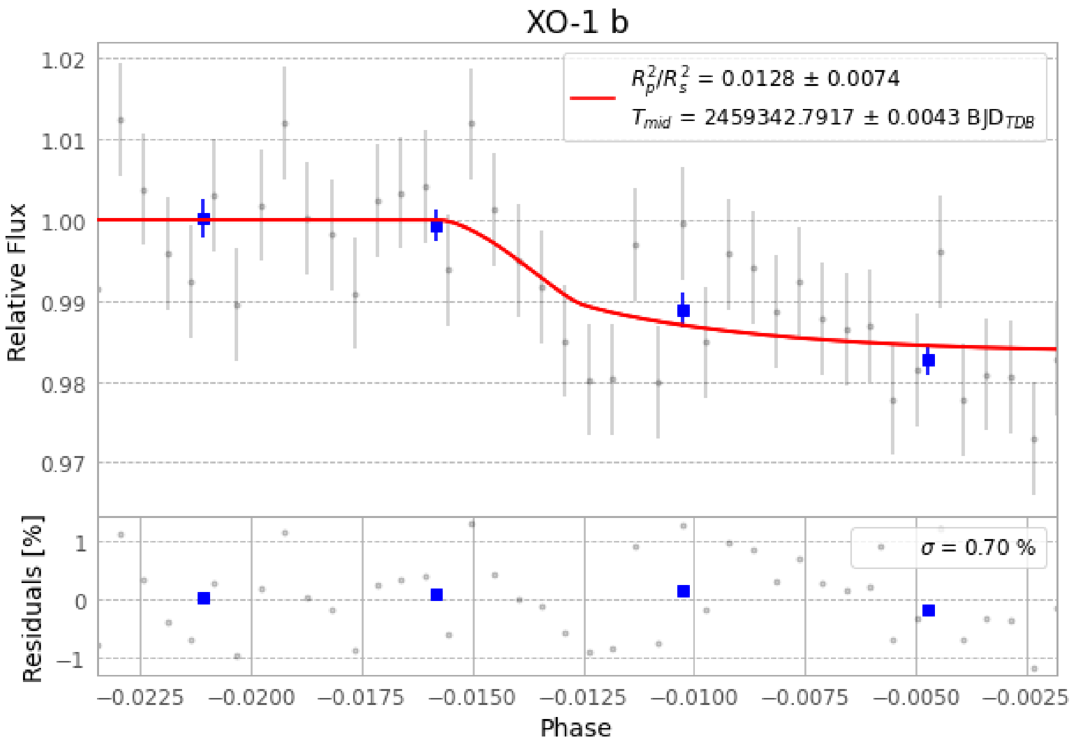

Figure 1 — FinalLightCurve_XO-1 b_2021-05-11.png (SHA-256 32fbb311883779ca…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 223], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 141ef630df591e95…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

42 FITS frames (28 MB), `XO-1210512031512.FITS` → `XO-1210512051812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-210512.log (f41dd028481b…), FinalLightCurve_XO-1 b_2021-05-11.png (32fbb3118837…), FinalLightCurve_XO-1 b_2021-05-11.csv (380e0cf5f9ef…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)